

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and validated **greenhouse gas emission models** for diverse agricultural systems, translating complex agronomic science into structured logic for integration into the VSAg platform.
- Collaborated with **VSAg Consultants software developers** to ensure accurate implementation, testing, and ongoing maintenance of agricultural emission models as the platform evolved.
- Researched and documented adherence to **IPCC methodologies** and national GHG inventory frameworks, contributing to the scientific credibility of the VSAg platform's carbon accounting.
- Authored detailed methodology notes and technical documentation explaining **complex scientific models** to both technical development teams and non-technical client stakeholders.
- Automated data ingestion and pre-processing for agricultural datasets using **Python scripting**, preparing inputs for climate modelling and analysis.
- Managed and analyzed large agricultural datasets using **SQL** across PostgreSQL and Oracle databases for model validation and reporting purposes.
- Containerized data processing and modelling components using **Docker**, ensuring consistent environments for development and deployment on **AWS EC2**.

SKILLS

Agricultural Science – Greenhouse Gas Modelling, Carbon Intelligence, Agricultural Systems, Tillage, Beef, Dairy, Pigs, Poultry, Mixed Farming, Agronomy

Climate & Sustainability – Scope 3 Emissions, Carbon Accounting, IPCC Methodologies, National GHG Inventory Frameworks, Climate Action

Data Analysis & Modelling – Quantitative Modelling, Statistical Analysis, Data Validation, Data Warehousing, Exploratory Data Analysis

Programming & Scripting – Python, R, SQL, pandas, NumPy, scikit-learn

Cloud & Development Tools – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins, Jira, Confluence

PROJECTS

GHG Emission Model Development for VSAg Platform

- Engineered Python scripts to process and clean diverse agricultural data sources, preparing inputs for greenhouse gas emission models.
- Developed structured logic for modelling methane and nitrous oxide emissions from dairy and beef farming systems, aligning with IPCC guidelines.
- Collaborated with VSAg Consultants to integrate model outputs into the platform's reporting dashboard, enabling clients to visualize their carbon footprint.

Agricultural Data Analysis & Reporting Dashboard

- Designed and implemented a data pipeline using Python and SQL to extract, transform, and load data from enterprise systems into a PostgreSQL data warehouse.
- Created interactive dashboards using **Power BI** and DAX measures to visualize key agricultural metrics and emission trends for internal stakeholders.
- Performed statistical analysis on historical yield and soil data using R to identify correlations relevant to carbon sequestration potentials.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)